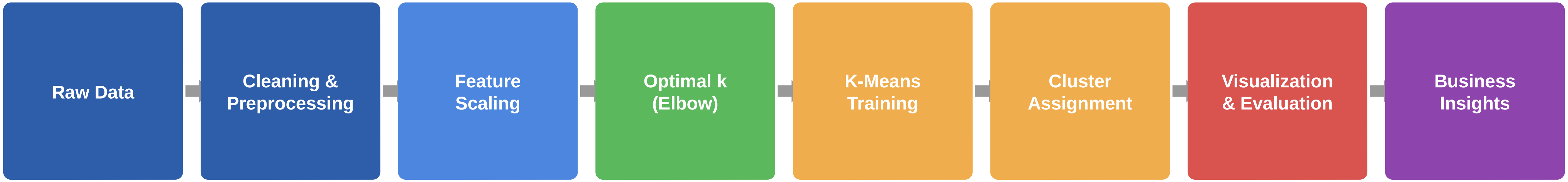
Customer Segmentation Analysis

Segmenting retail customers using K-Means Clustering

Objective

1. Segment customers into meaningful groups instead of treating them all the same
2. Identify the statistically optimal number of segments
3. Build a K-Means model that assigns new customers to the right segment
4. Translate clusters into business-friendly profiles for marketing use

Methodology



A sequential, reproducible pipeline: each stage's output feeds directly into the next — from raw customer data to actionable business insight.

Dataset

Source: Mall Customer Segmentation Dataset (Kaggle)

Records: 200 customers, no missing values

Format: Structured CSV

Split: 80% train (160) / 20% holdout (40)

Column	Type
CustomerID	Integer
Gender	Categorical
Age	Integer
Annual Income (k$)	Integer
Spending Score (1-100)	Integer

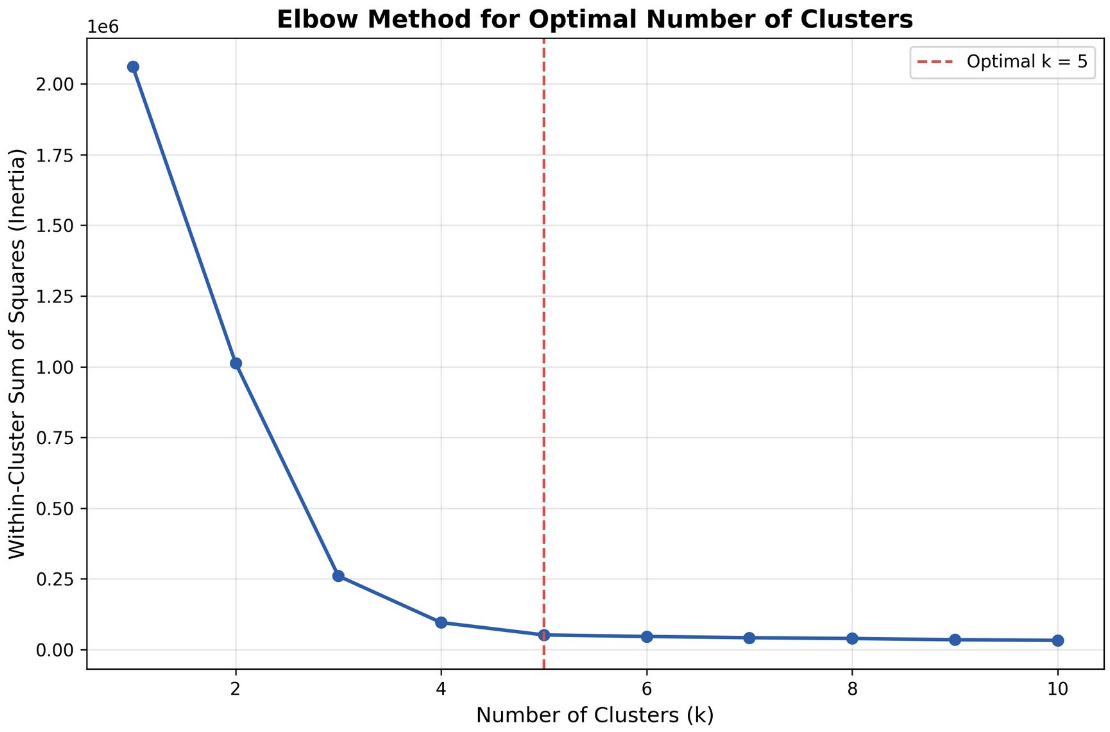
Algorithm

Technique

K-Means Clustering — partitions customers into k groups by minimizing within-cluster variance (WCSS). Hierarchical clustering used to validate structure.

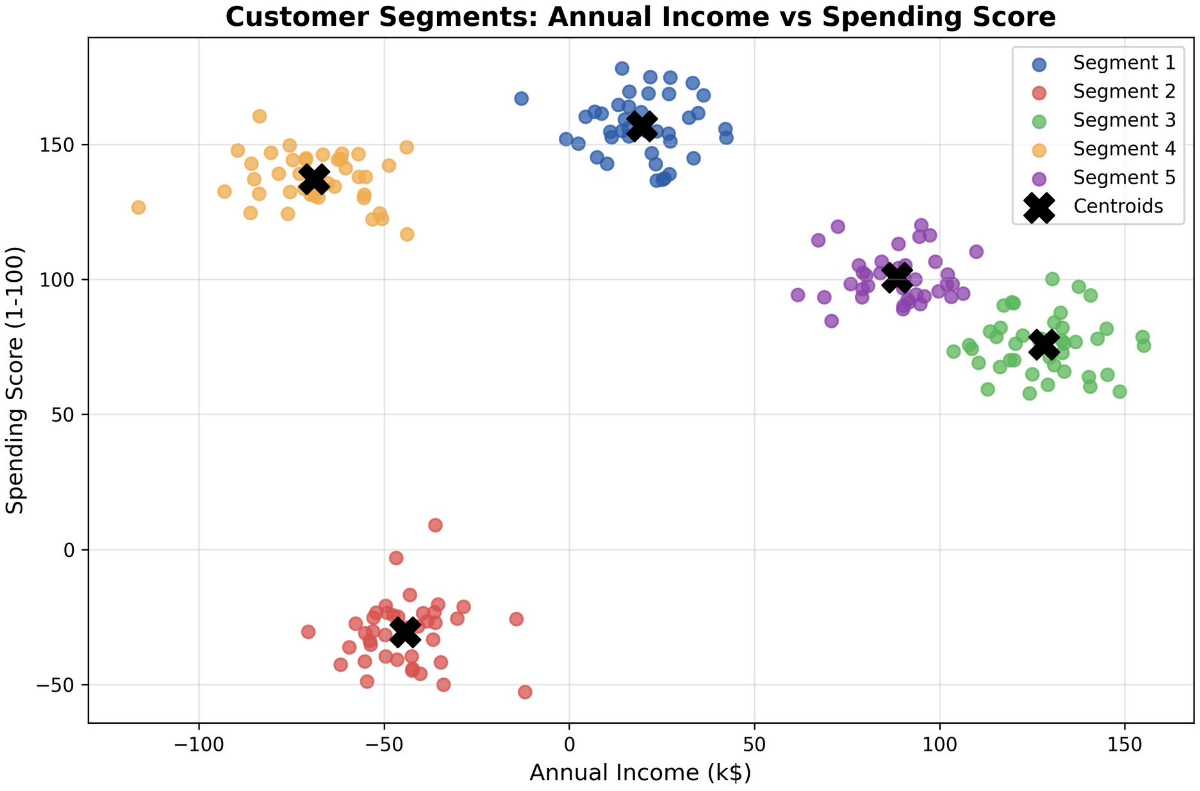
Optimal k Selection

Elbow Method (WCSS vs. k) + Silhouette Score — both point to k = 5 as optimal (silhouette ≈ 0.61).



Elbow Method: optimal k = 5

Results



Cluster	Profile	Count
0	High Income - Low Spend	41
1	High Income - High Spend	49
2	Low Income - High Spend	38
3	Low Income - Low Spend	38
4	Average Customers	34

Silhouette Score at k=5: ≈ 0.61 (well-separated clusters)

Conclusion

- K-Means successfully separated customers into 5 distinct, well-validated segments
- Clear business labels: High/Low Income × High/Low Spend, plus an Average group
- Enables targeted marketing, loyalty programs, and re-engagement campaigns
- Foundation for future work: real-time segmentation, predictive customer value